FitBuddy: AI Fitness Plan Generator using Gemini Models
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Project Testing Phase – FitBuddy

The **Project Testing Phase** focuses on checking whether the FitBuddy application works correctly, produces the expected AI-generated results, stores data properly, and responds correctly to user feedback.

1. Functional Testing

Test all major functions of the application:

- User details submission.
- Personalized 7-day workout-plan generation.
- Nutrition/recovery tip generation.
- Feedback submission.
- Workout-plan updating.
- User and plan storage.
- Admin view of users and plans.

2. AI Testing

Verify that:

- Gemini generates a structured **7-day workout plan**.
- The plan contains warm-up, main workout, and cooldown/recovery guidance.
- Nutrition tips are generated according to the selected fitness goal.
- User feedback correctly modifies the existing workout plan.

3. Database Testing

Check whether:

- User information is stored correctly.
- Original workout plans are saved.

- Updated workout plans are stored separately.
- Stored information can be retrieved correctly for the admin view.

4. Frontend Testing

Test the three main pages:

- **index.html** – Verify that all input fields and the Generate Plan button work correctly.
- **result.html** – Verify that the workout plan, nutrition tip, and feedback form are displayed correctly.
- **all_users.html** – Verify that user and plan information is displayed properly.

5. API and Route Testing

Test the main FastAPI routes:

- `/`
- `/generate-workout`
- `/submit-feedback`
- `/view-all-users`

FastAPI's interactive `/docs` interface can also be used to explore and test the API endpoints.

6. Deployment Testing

The application should be run locally using Uvicorn and accessed through the local browser interface. The deployment process should verify that the required dependencies, Gemini API configuration, frontend, backend, and database work together correctly.

7. Expected Testing Outcome

Input Validation → AI Response Testing → Database Testing → Feedback Testing → UI Testing → API Testing → Local Deployment Verification

The testing phase ensures that FitBuddy functions as intended and provides a complete flow from **user input to personalized workout generation, nutrition guidance, feedback-based plan updating, and data management**.

Continue testing FitBuddy

- Create FitBuddy test cases
- Define FitBuddy acceptance criteria